Warm-Up Quiz — Day 14

Projection Matrices, Two-Stage Algebra, and Ratios

Complete before lecture. 10 minutes.

Name: ______________________________

Today we study 2SLS, which projects the endogenous regressors onto the instrument space before running OLS. This quiz reviews projection matrices and the algebra of ratios that connects the Wald estimator to IV.

1. Projection Matrix Review.
Let $\boldsymbol{P}_Z = \boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'$ be the projection onto the column space of $\boldsymbol{Z}$.

(a) Verify that $\boldsymbol{P}_Z$ is idempotent: show $\boldsymbol{P}_Z^2 = \boldsymbol{P}_Z$.

(b) What is $\boldsymbol{P}_Z\boldsymbol{Z}$? (Simplify.)

(c) If $\boldsymbol{X} = \boldsymbol{Z}\boldsymbol{\Gamma} + \boldsymbol{V}$ (first-stage regression), what is $\boldsymbol{P}_Z\boldsymbol{X}$?

2. Two-Stage Algebra.
In 2SLS, Stage 1 produces $\hat{\boldsymbol{X}} = \boldsymbol{P}_Z\boldsymbol{X}$ and Stage 2 regresses $\boldsymbol{Y}$ on $\hat{\boldsymbol{X}}$:

$$\hat{\beta}_{2SLS} = (\hat{\boldsymbol{X}}'\hat{\boldsymbol{X}})^{-1}\hat{\boldsymbol{X}}'\boldsymbol{Y}$$

(a) Substitute $\hat{\boldsymbol{X}} = \boldsymbol{P}_Z\boldsymbol{X}$ and use $\boldsymbol{P}_Z^2 = \boldsymbol{P}_Z$ to simplify $\hat{\boldsymbol{X}}'\hat{\boldsymbol{X}}$.

(b) Similarly simplify $\hat{\boldsymbol{X}}'\boldsymbol{Y}$. Write the full 2SLS formula in terms of $\boldsymbol{X}$, $\boldsymbol{Y}$, and $\boldsymbol{P}_Z$.

3. The Wald Estimator as a Ratio.
With a binary instrument $Z_i \in \{0, 1\}$, the Wald estimator is:

$$\hat{\beta}_{Wald} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

where $\bar{Y}_1$ = mean of Y when $Z = 1$, etc.

(a) The numerator estimates the "reduced form" effect of Z on Y. In terms of the structural model $Y = \beta X + e$ and first stage $X = \pi Z + v$, what population quantity does $\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]$ equal?
(Hint: substitute.)

(b) The denominator estimates the "first stage" effect of Z on X. What population quantity does $\mathbb{E}[X|Z = 1] - \mathbb{E}[X|Z = 0]$ equal?

(c) Take the ratio. What do you get?

4. Variance Inflation from IV.
A rule of thumb: $\text{Var}(\hat{\beta}_{IV}) \approx \text{Var}(\hat{\beta}_{OLS})/\rho^2_{XZ}$, where $\rho_{XZ} = \text{Corr}(X, Z)$.

(a) If $\rho_{XZ} = 0.3$, by what factor are IV standard errors larger than OLS?

(b) Why does a weak instrument (ρ_{XZ} small) make IV unreliable? (One sentence.)

Answer Key — Day 14

1. (a) $\boldsymbol{P}_Z^2 = \boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\underbrace{\boldsymbol{Z}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}}_{=\boldsymbol{I}}\boldsymbol{Z}' = \boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}' = \boldsymbol{P}_Z$.

(b) $\boldsymbol{P}_Z\boldsymbol{Z} = \boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{Z} = \boldsymbol{Z}$.

(c) $\boldsymbol{P}_Z\boldsymbol{X} = \boldsymbol{P}_Z(\boldsymbol{Z}\boldsymbol{\Gamma} + \boldsymbol{V}) = \boldsymbol{Z}\boldsymbol{\Gamma} + \boldsymbol{P}_Z\boldsymbol{V} = \hat{\boldsymbol{X}}$. If $\boldsymbol{V}$ is the population error (orthogonal to $\boldsymbol{Z}$), $\boldsymbol{P}_Z\boldsymbol{V} \approx \boldsymbol{0}$, so $\hat{\boldsymbol{X}} \approx \boldsymbol{Z}\boldsymbol{\Gamma}$: the first-stage fitted values.

2. (a) $\hat{\boldsymbol{X}}'\hat{\boldsymbol{X}} = \boldsymbol{X}'\boldsymbol{P}_Z'\boldsymbol{P}_Z\boldsymbol{X} = \boldsymbol{X}'\boldsymbol{P}_Z\boldsymbol{X}$ (using $\boldsymbol{P}_Z' = \boldsymbol{P}_Z$ and $\boldsymbol{P}_Z^2 = \boldsymbol{P}_Z$).

(b) $\hat{\boldsymbol{X}}'\boldsymbol{Y} = \boldsymbol{X}'\boldsymbol{P}_Z\boldsymbol{Y}$. So $\hat{\beta}_{2SLS} = (\boldsymbol{X}'\boldsymbol{P}_Z\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{P}_Z\boldsymbol{Y}$.

3. (a) $\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0] = \beta(\mathbb{E}[X|Z=1] - \mathbb{E}[X|Z=0]) = \beta\pi$.

(b) $\mathbb{E}[X|Z=1] - \mathbb{E}[X|Z=0] = \pi$.

(c) $\beta\pi/\pi = \beta$. The reduced form divided by the first stage recovers the structural parameter.

4. (a) SE ratio $= 1/|\rho_{XZ}| = 1/0.3 \approx 3.3\times$ larger.

(b) A weak instrument inflates the variance of $\hat{\beta}_{IV}$, making confidence intervals very wide, and in finite samples the IV estimate is biased toward the OLS estimate.